DATA SCIENCE V1.0

Lesson 3

Documentation: Data Science V1.0

Introduction to Machine Learning

Objectives

- Understand ML theory foundations
- Implement classical algorithms
- Evaluate models properly
- Avoid overfitting

Topics

- AI vs ML vs DL vs GenAI
- Supervised learning
- Loss/Cost functions
- Train/test split
- Evaluation metrics:
    - Accuracy
    - Precision
    - Recall

    - F1
    - ROC-AUC
- Evaluation metrics:
    - For classification: accuracy, precision, recall, F1, ROC-AUC
    - For regression: MAE, MAPE, WAPE, MSE, RMSE.
- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Gradient Boosting
- AI History
- Data Splitting

Resources

- Scikit-learn documentation:
    - https://scikit-learn.org/stable/
- StatQuest (YouTube):
    - https://www.youtube.com/c/joshstarmer
- Google ML Crash Course:
    - https://developers.google.com/machine-learning/crash-course

Practice Project – Customer Churn Prediction System
Requirements

1. Preprocess dataset
2. Train:
    - Logistic Regression
    - Decision Tree
    - Random Forest
3. Compare models
4. Evaluate:
    - For classification models:
        a. Confusion matrix
        b. ROC curve
        c. Precision/Recall tradeoff
    - For regression models:
        a. MAE,
        b. MAPE,
        c. MSE
5. Feature importance analysis

Advanced Requirement

- Show one case of overfitting
- Apply regularization
- Improve performance

Deliverables

- Notebook
- Model comparison table
- Business explanation:
    - Why this model?
    - Interpretation of model performance (what do metrics tell us about the model)